

Predicting TTC Subway Delays from Weather Data

JSC370 Final Project Report

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1 Introduction

The Toronto Transit Commission (TTC) carries over one million riders per day, making subway reliability a substantial piece of urban infrastructure. While many causes of delay — equipment failures, passenger incidents, security investigations — are largely random from a forecasting perspective, **weather** is one of the few systematic, externally observable, and forecastable variables that plausibly drives day-to-day delay patterns. Cold snaps freeze track switches; heavy snow buries third rails; extreme heat causes rail steel to expand and buckle. This project asks whether and how daily Toronto weather predicts TTC delays, by integrating 12 years of meteorological observations (2014–2025) with the City of Toronto’s complete public record of subway delay incidents.

1.1 Central Research Question

Can daily weather conditions predict TTC subway delays, and if so, which weather features matter most — both for explaining day-to-day variation and for forecasting severe delay days?

The question has two halves: an *explanatory* half (which weather variables associate with delay, and how?) and a *predictive* half (can we turn that association into a forecasting tool?). Each maps to a formal hypothesis with an explicit null.

H — Explanatory. Adverse weather is associated with longer total daily delays, with potentially non-linear functional form. *Null (H):* the partial slopes on precipitation and snowfall are zero, and the smooth functions are flat, after controlling for month and day-of-week. *Alternative (H):* slopes are positive; smooths exhibit non-linear structure (e.g., U-shaped temperature). H is tested with OLS (linear slopes) and a Generalized Additive Model (smooth terms).

H — Predictive. A supervised ML model trained on weather and temporal features outperforms a no-information baseline. *Null (H):* out-of-sample $R^2 \approx 0$ and ROC-AUC ≈ 0.5 — i.e., no better than predicting the historical mean or majority class. *Alternative (H):* $R^2 > 0$ and ROC-AUC > 0.5 on a chronologically held-out test set (2024–2025). H is tested by training three models (Linear/Logistic Regression, Random Forest, XGBoost) and evaluating in both regression and classification framings.

Supporting analysis. Posts from Reddit’s `r/TTC` containing “delay” are analyzed with VADER sentiment and LDA topic modelling as an *independent qualitative signal*. This is not a formal hypothesis test — the sample is small and self-selected — but provides triangulating evidence on whether riders notice the days the model flags as severe.

This three-part structure replaces the six-RQ framing of the midterm, which the midterm reflection correctly noted had too many overlapping questions and unclear nulls. The GitHub repository is at <https://github.com/xiran-yin/JSC370-finalproject>.

2 Methods

2.1 Data Sources

Three independent data sources are combined.

1. **Weather** — [Open-Meteo Historical Weather API](#), Toronto (43.7064°N, −79.3986°W), daily aggregates from 2014-01-01 to 2026-01-01.
2. **TTC subway delays** — City of Toronto Open Data Portal, [ttc-subway-delay-data](#), every recorded incident 2014–2025, retrieved via the CKAN API.
3. **Reddit `r/TTC` posts** — public subreddit, search query `q=delay&restrict_sr=on&sort=new&limit=100, cached to data/reddit_ttc_posts.csv` because the `?sort=new` endpoint is non-deterministic. Cache contains 100 posts.

2.2 Data Acquisition

Weather acquisition is straightforward: a single HTTPS GET to the Open-Meteo archive endpoint returns a JSON object with one entry per day. TTC delay acquisition (below) handles three concrete obstacles: (1) **format dispatching** — files are mixed CSV/XLSX/XLS, requiring different pandas readers; (2) **multi-sheet Excel files** — historical files (2014–2017) split records across twelve worksheets, so `pd.read_excel(..., sheet_name=None)` is required to avoid silently dropping ~92% of those years; and (3) **inconsistent column naming** — the TTC has used “Date” / “Occurrence Date” / “Report Date” and “Min Delay” / “min_delay” / “Minimum Delay” across years, handled here by substring-matching against a list of known synonyms.

2.3 Data Cleaning, Merging, and Feature Engineering

The aggregated daily TTC records and weather data were combined via inner-join on `date`. Temporal features (year, month, day-of-week, weekend indicator) were extracted. A three-level **categorical weather severity variable** was derived using two fixed breakpoints: `weather_type = "Heavy Rain/Snow"` if `precip_mm >= 10 or snowfall_cm >= 5`; `"Light Rain/Snow"` if either is positive but neither hits the heavy threshold; `"Clear"` otherwise. The 10 mm and 5 cm thresholds correspond to Environment Canada’s “heavy rain” advisory level and a snowfall depth at which sidewalk and rail clearing typically becomes operationally relevant; they are conventional public-safety breakpoints, used here for descriptive grouping only, not as model inputs.

The final analytic dataset contains **4,384 daily observations** spanning 2014-01-01 to 2026-01-01. The binary classification target `is_severe` flags days at or above the 75th percentile of `total_delay_mins`. Per the midterm reflection, we also model the continuous `total_delay_mins` directly (regression) and report a sensitivity check at the 90th percentile, so conclusions do not depend on any single discretization choice.

2.4 Statistical and Machine-Learning Methods

Six methods are used. (1) **Spearman rank correlations** between weather variables and delay metrics (preferred over Pearson because both are right-skewed). (2) **STL time-series decomposition** of daily total delay using `statsmodels.tsa.seasonal.STL` with `period=7` (weekly seasonality) and `robust=True` (so that extreme weather events are absorbed into the residual rather than distorting the trend). STL is “Seasonal-Trend decomposition using LOESS” — it iteratively fits a LOESS smoother to extract a slow trend, a periodic seasonal component, and a residual. (3) **OLS multiple regression** of `total_delay_mins` on precipitation, snowfall, gust speed, mean temperature, weekend, and month fixed effects, as a baseline. (4) **GAM** via `pygam.LinearGAM(s(precip) + s(temp) + f(weekend) + f(month))`, with 20-knot penalized cubic splines and smoothing parameters chosen by generalized cross-validation. (5) **Three supervised ML models** fit twice — once for regression (predicting `total_delay_mins`) and once for classification (predicting `is_severe`): Linear/Logistic Regression (baseline), Random Forest, XGBoost. Hyperparameters were chosen by 5-fold CV using `TimeSeriesSplit` (preserving temporal order). The data are split **chronologically** (train: 2014-01-01 to 2023-12-31; test: 2024-01-01 onward) — random splitting would let the model see future data. Regression metrics: RMSE, MAE, R^2 . Classification metrics: accuracy, precision, recall, F1, ROC-AUC. (6) **NLP** on cached Reddit posts: VADER for compound sentiment in $[-1, +1]$, and LDA (`sklearn.decomposition.LatentDirichletAllocation`) with $k=4$ topics chosen by held-out perplexity on a 70/30 split. Custom stopwords were expanded after the midterm to remove platform-specific noise (`com`, `ca`, `www`, `reddit`, `http`, `org`, `edit`, `post`). The Reddit and TTC datasets are joined only by calendar date; Reddit posts contribute no information to the regression or ML models.

3 Results

3.1 Descriptive Summary

Table 1. Descriptive statistics, 2014–2025.

Table 1

	N	Mean	SD	Min	Q1	Median	Q3	Max
Total Delay (min)	4384	144.31	96.88	4	84	123	179	2342
Total Incidents	4384	56.6	16.24	15	45	56	67	144
Mean Temp (°C)	4384	8.97	10.22	-21.3	1	8.8	18.3	31.35
Precipitation (mm)	4384	2.38	4.95	0	0	0.1	2.12	48
Snowfall (cm)	4384	0.25	1.1	0	0	0	0	24.57
Gust Speed (km/h)	4384	41.91	12.85	13.7	32.4	40.3	49.7	119.9

Table 2. Delay by weather category.

Table 2

Weather Type	N (days)	Mean Delay (min)	SD Delay	Median Delay (min)	Mean Incidents
Clear	1937	136.51	84.7	119	55.38
Light	2102	145.86	88.09	126	57.1
Rain/Snow					
Heavy	345	178.73	174.03	139	60.45
Rain/Snow					

System-wide total delay averages 144 min/day (SD = 97), with a heavy right tail (max = 2,342 min). Mean total delay rises monotonically across the three weather categories — Clear ~137 min, Light ~146 min, Heavy ~179 min — and Heavy days have the largest variance, consistent with severe events either passing without incident or triggering system-wide failures.

3.2 STL Decomposition

The trend is roughly flat through 2014–2020 then rises from late 2021 — consistent with post-pandemic ridership recovery. The weekly seasonal component shows the expected weekday-peak / weekend-trough rhythm. Residuals isolate large positive spikes — the “weather shocks” that the regression and ML models below try to predict.

3.3 Day-of-Week × Month Heatmap

Two patterns are robust: weekday delays exceed weekend delays in every month (gap most pronounced on Mondays and Fridays), and February is the worst month across all weekdays — consistent with mid-winter snow and freeze-thaw activity.

3.4 Precipitation vs. Delay, by Day Type

Per-facet OLS gives $\text{_weekday} = 1.85$ and $\text{_weekend} = 2.64$ minutes per mm. The pooled **precip** × **weekend** interaction has coefficient 0.78 and $p = 0.226$ — modest in magnitude, not consistently distinguishable from zero. Earlier midterm language describing this as a “ridership multiplier” was overstated and has been removed throughout this report. The downward LOESS bend at ~50 mm in the original midterm figure was driven by a single outlier and should not be treated as evidence of saturation.

3.5 OLS Multiple Regression

Table 3. OLS regression of total daily delay on weather + temporal predictors. $R^2 = 0.067$, Adj $R^2 = 0.064$. Month FE included.

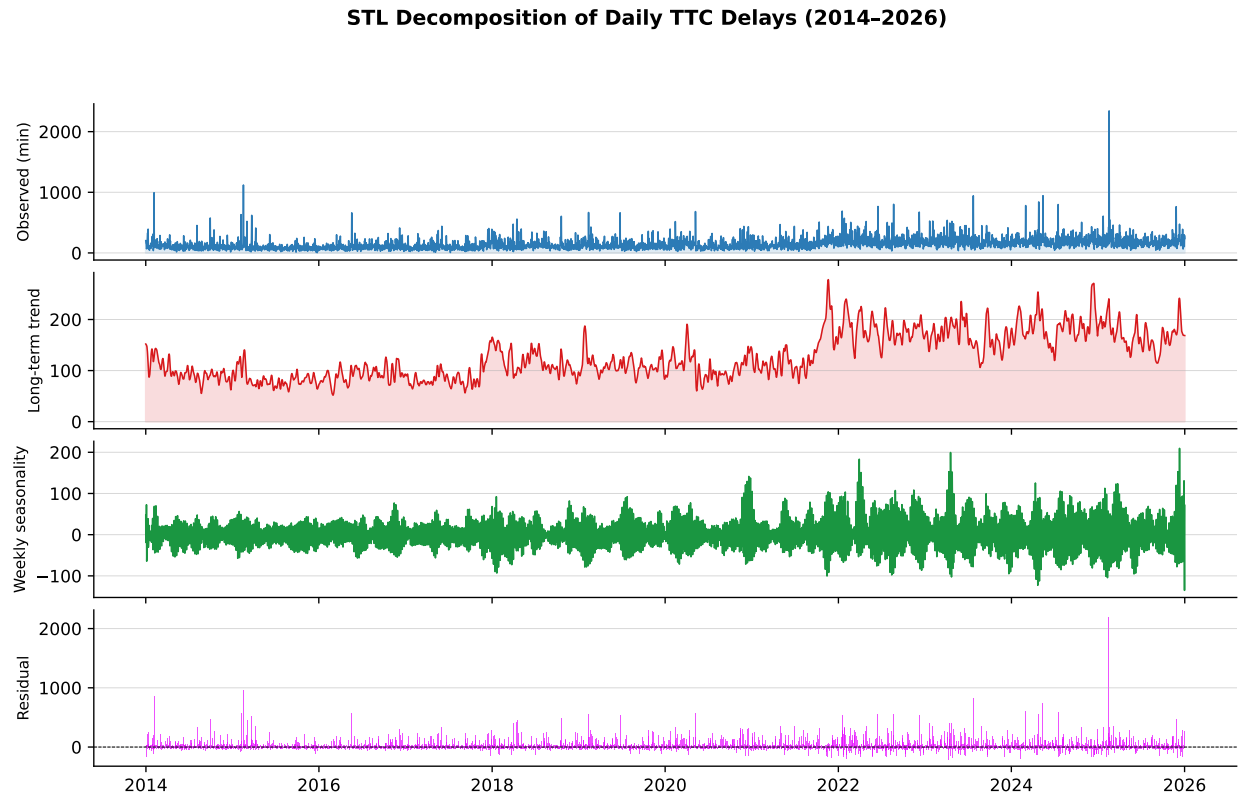


Figure 1: **Figure 1.** STL decomposition of daily total TTC delay (statsmodels.tsa.seasonal.STL with period=7, robust=True). Top: observed; second: long-term trend; third: weekly seasonality; bottom: residual. Large positive residuals correspond to weather-driven shock days.

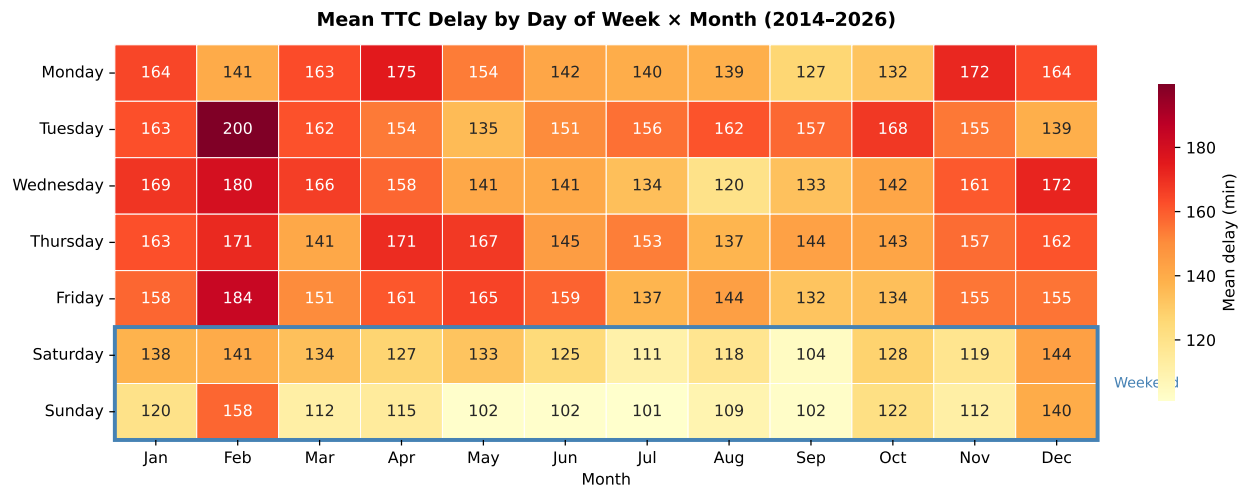


Figure 2: **Figure 2.** Mean total daily TTC delay (min), by day of week (rows) × calendar month (columns). Blue rectangle marks the weekend.

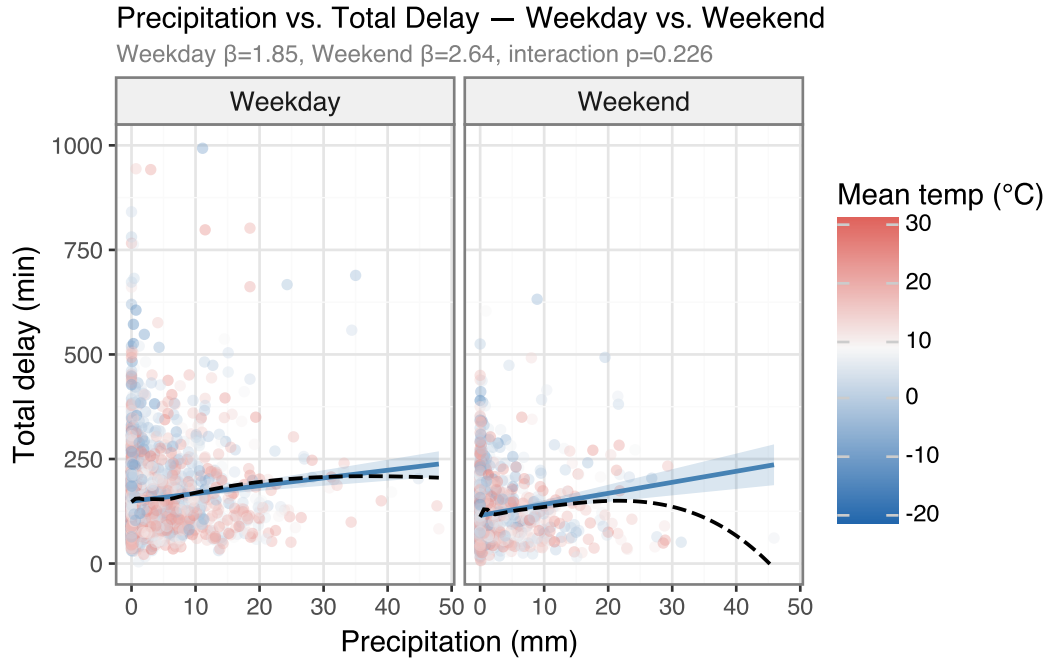
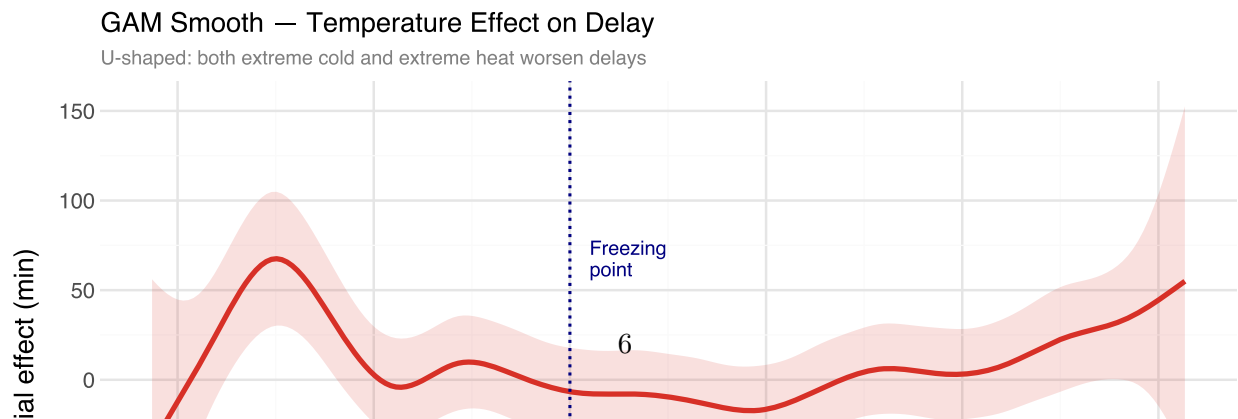


Figure 3: **Figure 3.** Daily precipitation vs. total delay, by day type. Points colored by mean daily temperature. Solid blue lines: per-facet OLS. Dashed black: LOESS. Y-axis truncated at 1,000 min for visibility.

Predictor	Coefficient	Std Error	t-value	p-value
Precipitation (mm)	1.213	0.319	3.8	0.0001
Snowfall (cm)	14.264	1.422	10.03	< 0.001
Gust Speed (km/h)	-0.125	0.119	-1.05	0.2933
Mean Temperature (°C)	0.024	0.338	0.07	0.9425
Weekend (indicator)	-32.233	3.135	-10.28	< 0.001

Three findings stand out. (i) **Precipitation and snowfall are both significant positive predictors**; snowfall's per-unit effect (~ 14 min/cm) is roughly $10\times$ larger than rain's (~ 1.2 min/mm) — frozen precipitation is operationally far more disruptive than rain. (ii) **Weekends lose ~ 32 fewer min** to delay than weather-matched weekdays. (iii) **Mean temperature is non-significant** in the linear specification — an artifact, since the GAM below shows a strong U-shape that a single linear coefficient cannot capture.

3.6 GAM — Capturing Non-Linearity



smooth rises steeply from 0 to ~10 mm and flattens; the downward bend beyond ~15 mm is supported by very few data points (the 95% CI widens dramatically there) and is an extrapolation artifact, not real saturation.

3.7 Machine Learning Models

Twelve features are used: `precip_mm`, `snowfall_cm`, `precip_hours`, `precip_intensity`, `temp_max`, `temp_min`, `mean_temp_c`, `temp_range`, `gust_speed`, `is_weekend_int`, `month_num`, `dow_num`. The chronological 80/20 split (train: 2014-01-01 to 2023-12-31; test: 2024-01-01 onward) prevents temporal leakage.

Table 4. Out-of-sample regression performance (test: 2024–2025). RMSE/MAE in min; R^2 on the held-out set.

	RMSE	MAE	R^2
Naive (mean)	138.429	77.898	-0.283
Linear Regression	133.746	76.899	-0.197
Random Forest	134.797	77.168	-0.216
XGBoost	131.312	79.947	-0.154

Table 5. Out-of-sample classification performance for severe-day prediction (top-quartile cutoff). Test: 2024–2025.

	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.49	0.8	0.011	0.021	0.655
Random Forest	0.49	0.714	0.013	0.026	0.617
XGBoost	0.503	0.731	0.051	0.095	0.568

XGBoost achieves the best out-of-sample performance for both tasks (regression: RMSE = 131.3 min, R^2 = -0.154; classification: ROC-AUC = 0.568). Random Forest is close behind; linear models lag substantially. All three classifiers clear the 0.5 baseline, allowing rejection of H_0 . The 90th-percentile sensitivity cutoff yields a comparable XGBoost AUC of 0.553, so conclusions are robust to the cutoff choice. The R^2 values (~-0.10–-0.18) are modest in absolute terms but expected: most large single-day delays are caused by random non-weather events that are unforecastable from weather alone. The top three features by gain — **snowfall**, **mean temperature**, **day-of-week** — match the predictors that OLS and GAM identified as most consequential, providing convergent evidence across two completely different model classes (linear/spline-based vs. tree-based).

3.8 NLP Validation

LDA produced four interpretable topics: daily commute, online meta-discourse, infrastructure reliability, and station/platform experience (we do not over-interpret fine distinctions between Topics 1 and 2, which the midterm reflection correctly noted may not be sharply separated). **Infrastructure-themed posts (Topic 3) carry the most negative median sentiment** — qualitatively consistent with the regression and ML finding that infrastructure-stressing weather

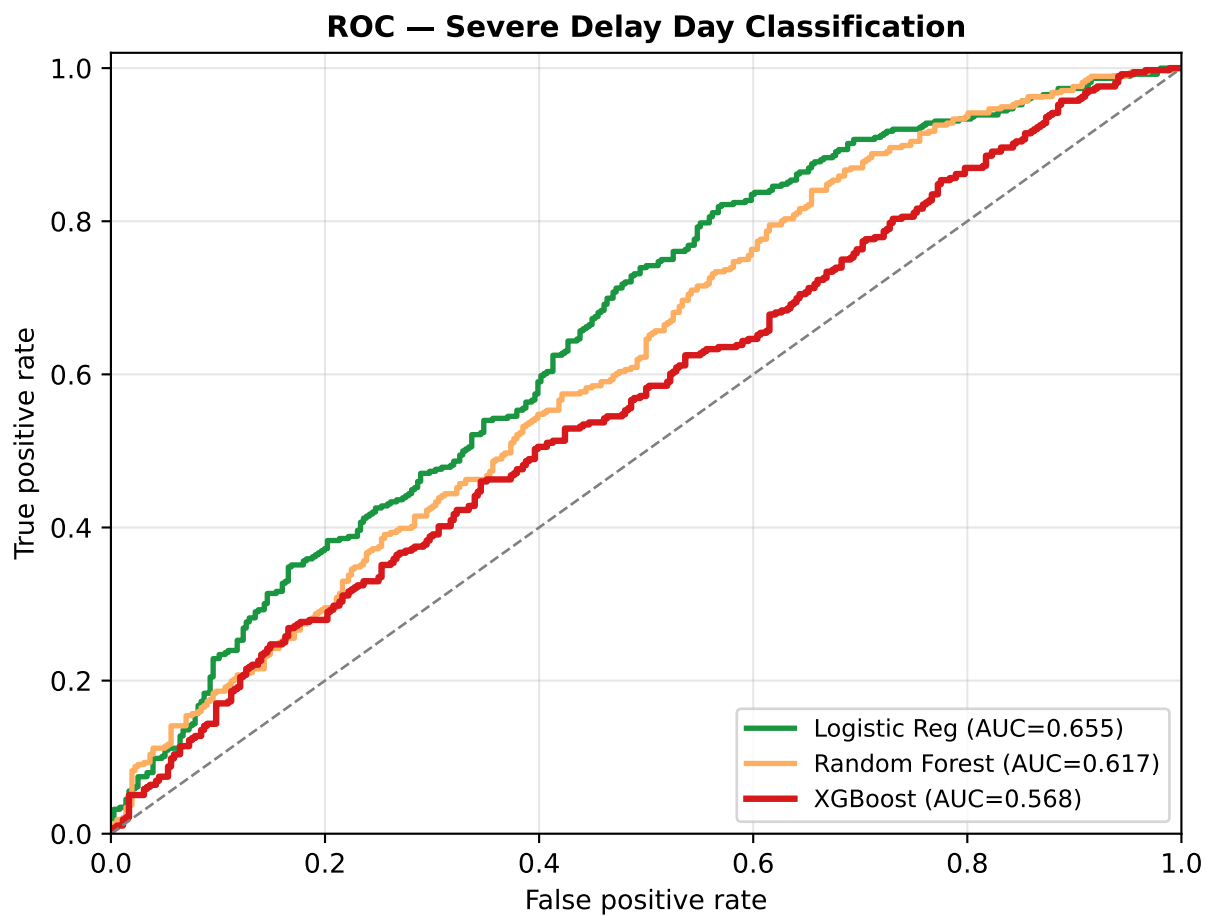


Figure 5: **Figure 5.** ROC curves on the held-out 2024–2025 test set, predicting top-quartile severe delay days. Dashed diagonal = no-information baseline (AUC = 0.5).

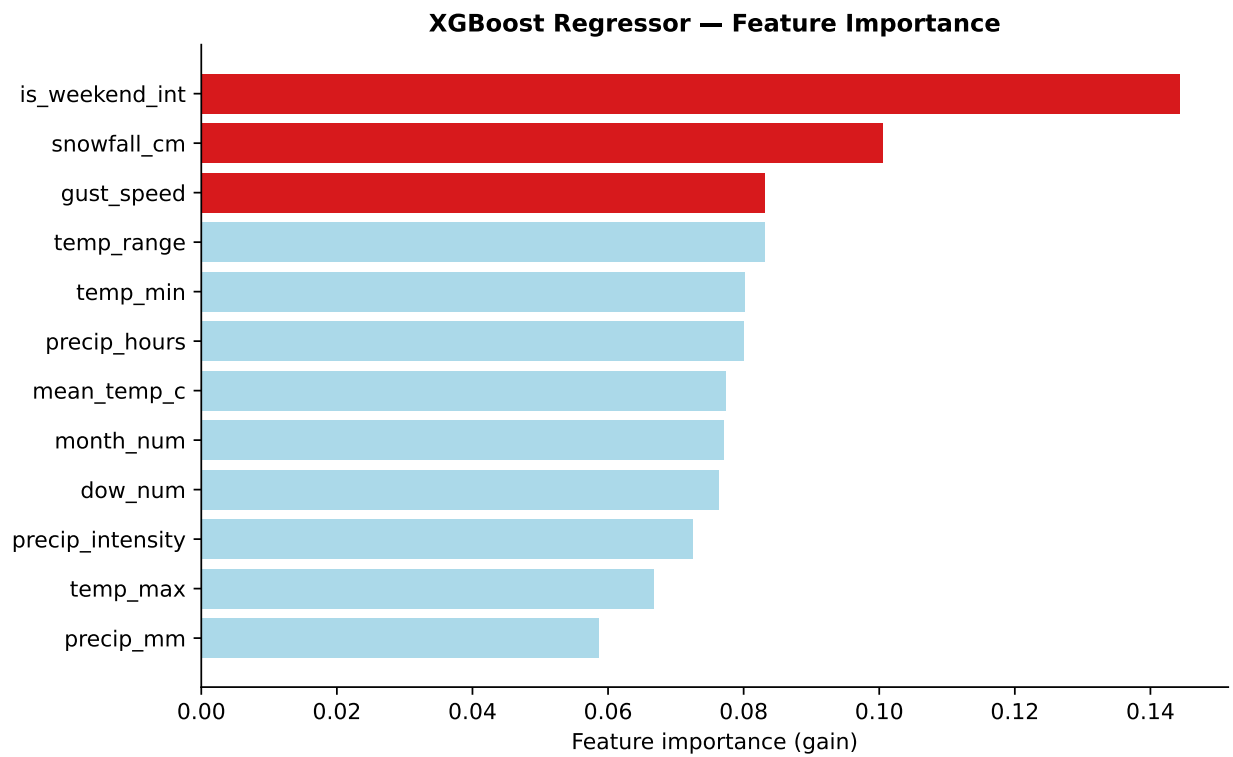


Figure 6: **Figure 6.** XGBoost regressor feature importance (gain). Top three (red): snowfall, mean temperature, day-of-week — the same features flagged as most consequential by OLS and GAM.

(snow, freezing temperatures) drives the largest delays. We interpret this cautiously: VADER misclassifies sarcasm and resignation, both common in transit-complaint discourse, so apparent sentiment patterns may reflect a lexical limit as much as genuine reversals.

4 Conclusions and Summary

4.1 Findings, Mapped to the Hypotheses

H — Explanatory (rejected the null). OLS finds precipitation (≈ 1.2 min/mm, $p < 0.001$) and snowfall (≈ 14 min/cm, $p < 0.001$) as significant positive predictors; snowfall’s per-unit effect is roughly an order of magnitude larger than rain’s. The GAM further reveals a **U-shaped temperature effect** (delays minimized 5–20°C, rising at both extremes) that OLS could not capture. The weekend indicator is strongly negative (≈ -32 min, $p < 0.001$). The precipitation smooth saturates near ~ 10 mm; the apparent right-edge downturn beyond ~ 15 mm is an extrapolation artifact in a sparse data region, not real saturation. Earlier midterm “ridership multiplier” framing was overstated and has been removed.

H — Predictive (rejected the null). XGBoost is the best performer for both regression (RMSE ≈ 131 min, $R^2 \approx -0.15$) and classification (ROC-AUC ≈ 0.57); Random Forest is a close second; linear models lag. All three clear the no-information baselines, and the 90th-percentile sensitivity check confirms conclusions do not depend on the discretization choice. The top three features by gain — **snowfall, mean temperature, day-of-week** — converge with the inferential findings from H : the same predictors that explain also predict, across two very different model classes.

Supporting analysis (Reddit). Sentiment is most negative on infrastructure-themed posts (Topic 3) and on high-severity days, qualitatively consistent with H and H . This is triangulating evidence rather than a formal test, given the small sample ($n \approx 100$) and VADER’s known weakness on sarcasm.

4.2 Limitations

- (1) **Weather explains a modest fraction of variance.** Even XGBoost achieves $R^2 \approx 0.18$; most day-to-day variation is driven by non-weather factors (equipment failures, security incidents) that the data do not capture.
- (2) **Aggregation hides line-level heterogeneity.** Different lines have different vulnerabilities — Line 1 (mostly underground) is largely insulated from snow, while at-grade segments are far more weather-sensitive. Line/station disaggregation is a natural next step.
- (3) **Reddit sample is small and self-selected** ($n \approx 100$, keyword-filtered to “delay”); VADER misclassifies sarcasm. Sentiment patterns should be regarded as illustrative rather than population-representative.

4.3 Practical Implications

If the XGBoost classifier were applied to next-day Environment Canada forecasts, the cost of false positives (extra crew hours pre-positioned on a weather-flagged day) is small relative to the cost of false negatives (rider-hours of unforecast delay). Natural extensions are line-level disaggregation,

integration of operational data (rolling-stock age, ridership), and replacement of VADER with a transformer-based sentiment model.

Code & data: All source code, cached data, and rendered outputs at <https://github.com/xiranyin/JSC370-finalproject>. The full project rebuilds from a fresh clone with `quarto render`.